Self-Reconfigurable Soft Robotic Modules: Undergraduate Student Researcher

Rowan University · May 2024 – Present

Self-reconfigurable soft robotic modules require compact, reliable, and modular electronics to drive actuators, interface with sensors, and communicate between units. The challenge is designing PCBs that integrate seamlessly with soft robotic hardware under strict space and reliability constraints, including the ability to maintain flexion during operation. I own the module electronics: the custom flexible mainboards, the docking and addressing architecture, and the on-board actuation drivers, now in their fourth design iteration.

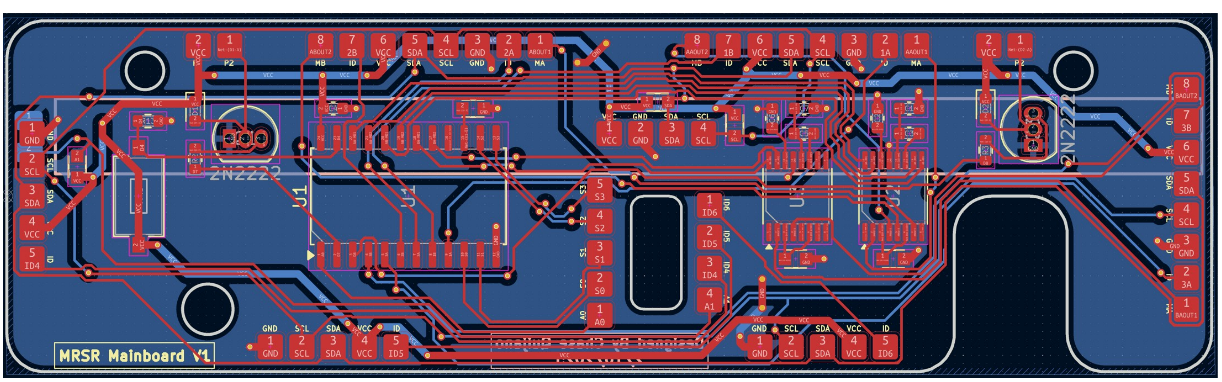
- **Iterated four generations of custom flexible PCBs:** engineered to withstand extreme mechanical strain and high-angle bending without trace de-lamination.
- **Integrated dynamic I^2C multiplexing and structural addressing onto individual docking faces:** enabling interlocking actuator modules to autonomously discover and map neighboring orientation nodes.
- **Embedded localized motor drivers and transistor-driven solenoid arrays on-board:** managing autonomous module interlocking and soft-actuator inflation cycles, supporting one forthcoming co-authorship.

Design Constraints

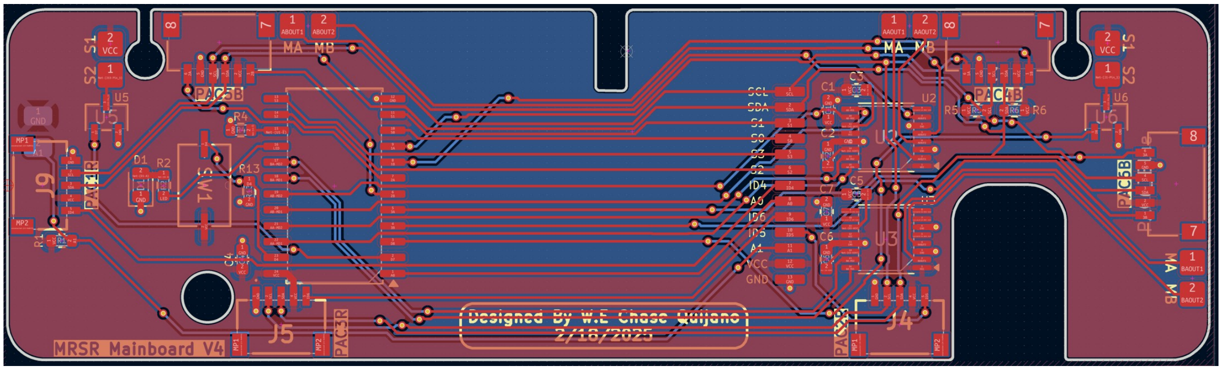
- The board must fit inside each soft robotic module: a 96×28 mm outline with mechanical keep-outs and a low-profile component height budget.
- Flexible PCB capable of maintaining roughly 1° of flexion for extended periods without trace failure.
- Actuator control and sensor input on-board, including a bend sensor for future positional data collection.
- Reliable power distribution and communication between modules.
- Two-layer layouts in KiCad to stay within the research budget, manufacturable with flexible PCB processes and assembled in the lab; low-profile SMD components throughout, with DRV8833 motor drivers selected for reliability, availability, and current capacity matched to the actuators.

Four Mainboard Generations

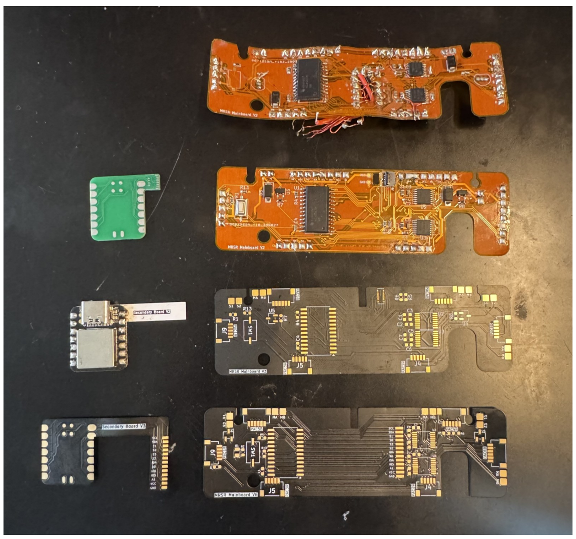
The mainboard is now on its fourth iteration, and the progression is the clearest record of my growth as an electronics designer. The first board shipped with a few incorrect traces that had to be bridged with jumper wires, and used through-hole transistors that fought the space constraints. Each generation after that folded in the lessons of the last: verifying every net through design review before fabrication, moving fully to SMD to shrink the footprint and simplify assembly, replacing discrete microcontroller interface pads with a compact 11-pin FPC connector to a secondary microcontroller board, and consolidating the routing for reliability under repeated flexion. Comparing the V1 and V4 layouts below shows how much the design matured: cleaner net organization, deliberate component placement, and a board designed to be manufactured and serviced, not just to work once.



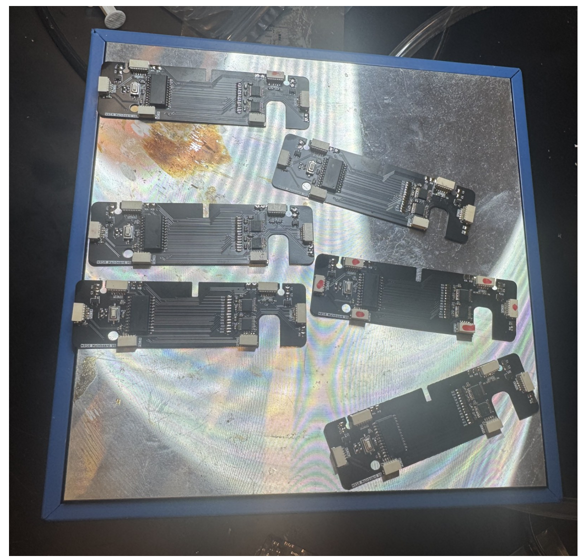
Mainboard V1 layout: first-generation two-layer design with through-hole transistors and discrete interface pads. This revision required jumper-wire fixes after fabrication.



Mainboard V4 layout: the latest iteration, fully SMD with FPC interconnects, consolidated routing, and placement driven by the module's mechanical and flexion constraints.



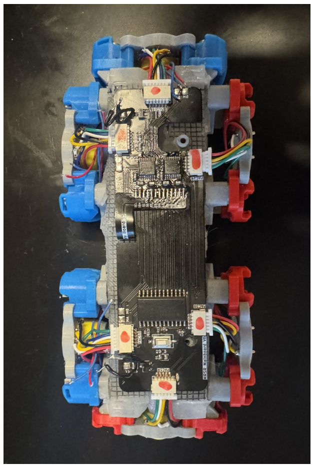
The mainboard lineage laid out: early flexible generations with hand-bodged fixes alongside the clean later revisions and their secondary microcontroller boards.



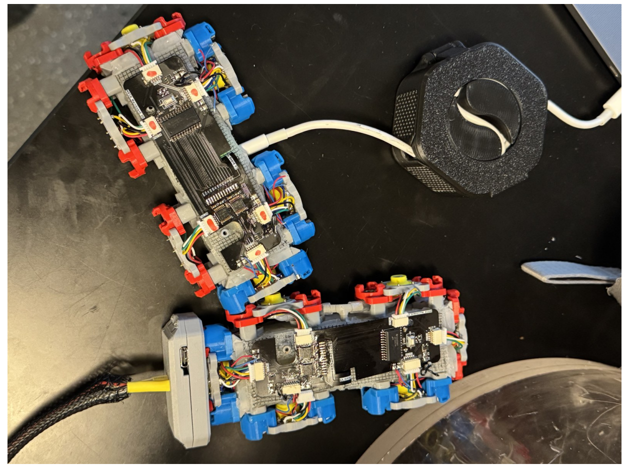
A production run of the latest mainboard revision, hand-assembled and ready for module integration.

Autonomous Docking and On-Board Actuation

Each docking face carries its own addressing hardware: dynamic I^2C multiplexing and structural addressing let a module identify which neighbor is attached to which face and in what orientation, so a chain of interlocked modules can autonomously discover and map its own topology. The mainboard also embeds the actuation layer locally, with motor drivers and transistor-driven solenoid arrays managing the interlocking mechanisms and soft-actuator inflation cycles without external drive electronics.



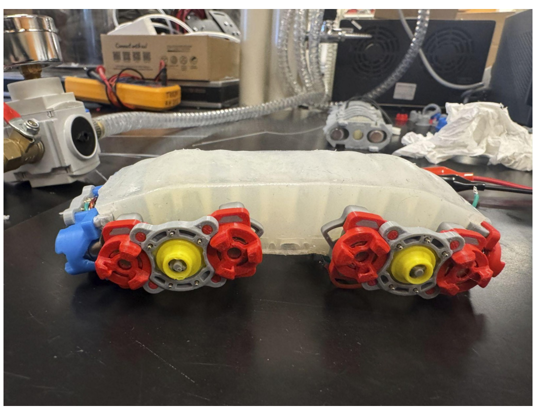
Assembled module with the latest mainboard installed: docking faces, PAC connectors, and actuation wiring integrated around the flexible board.



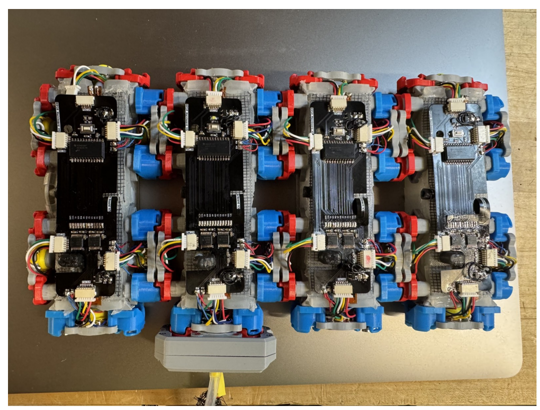
Two modules interlocked and powered over USB during intermodule communication bring-up.

Assembly, Testing, and Validation

All boards are hand-assembled in the lab using SMT techniques (hot plate and heat gun with solder paste) and inspected under a microscope for solder joint integrity and trace bridging. Electrical validation confirmed continuity and power integrity, and assembled modules have successfully powered their actuators under load with stable performance, including soft-actuator inflation testing. Extended flexion-cycle reliability testing is planned in the upcoming research phase.



Module inflation test: the soft actuator pressurized with the docking hardware attached.



Four modules with the latest mainboard revision installed, top-down view.

Collaboration and Research Impact

I work alongside graduate researcher Joshua Knospler to keep the PCB form factor matched to the module mechanical design, and I assist in module assembly and hardware refinements. The PCB systems described here extend the group's published soft-robotics foundations by enabling more compact, manufacturable, and modular electronics integration within the modules. This contribution is included in a forthcoming publication on which I am a co-author, and the research is currently being evaluated for potential patenting and commercialization.

Selected publications from this research group:

1. J. Knospler, W. Xue, and M. Trkov, "Reconfigurable modular soft robots with modulating stiffness and versatile task capabilities," *Smart Materials and Structures*, vol. 33, no. 6, 2024.
2. J. Knospler, N. Pagliocca, W. Xue, and M. Trkov, "TendrilBot: Modular Soft Robot with Versatile Radial Grasping and Locomotion Capabilities," *Sensors and Actuators A: Physical*, 2024.
3. J. Knospler, W. Xue, and M. Trkov, "MagBot: Reconfigurable Modular Soft Pneumatic Actuators with Tunable Magnetic Connection Mechanism," *IEEE AIM*, 2024.
4. J. Knospler, W. Xue, and M. Trkov, "A Shared Electrical-Pneumatic and Reversible Locking Intermodule Connector for Modular Robots," *IEEE AIM*, 2024.
5. J. Knospler, N. Pagliocca, W. Xue, and M. Trkov, "Realizing Modular Self-reconfiguring Soft Robots through Inter-module Communication and Model Checking," *IEEE RoboSoft*, 2025.